PANIMALAR ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University, Chennai

Department of Artificial Intelligence & Machine Learning

ADAPTIVE WIND POWER FORECASTING USING CONCEPT DRIFT DETECTION

A Self-Healing AI System for Renewable Energy Grid Reliability | Final-Year Capstone 2026

TEAM : B14

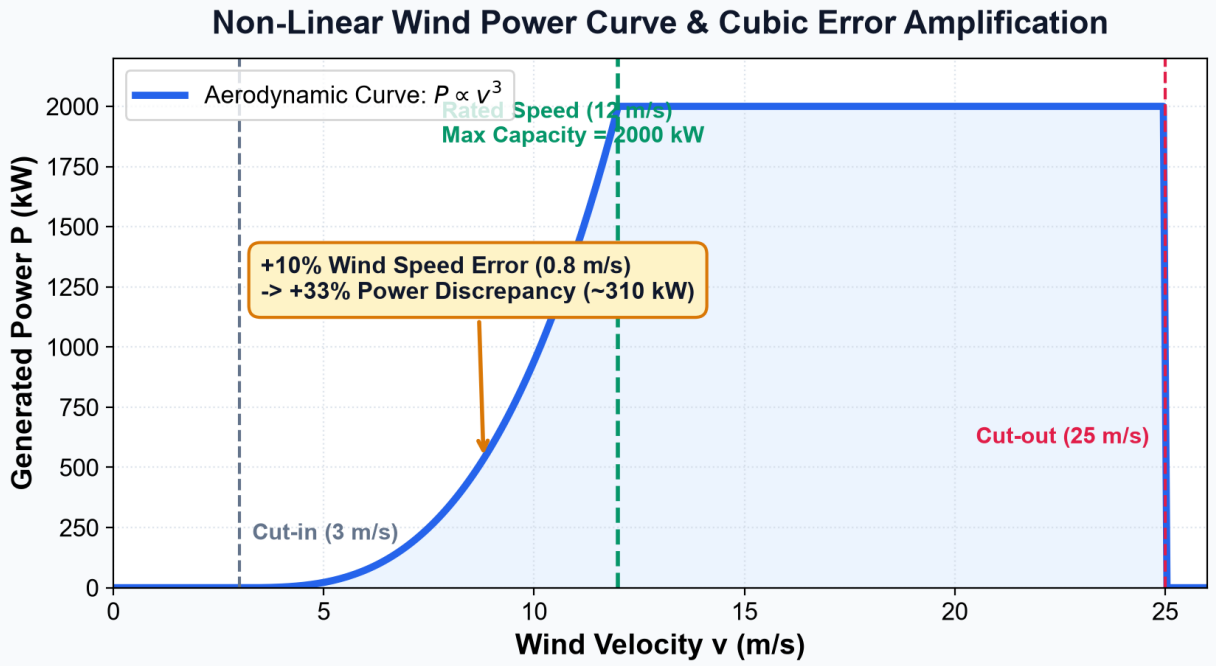
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Head of Department: Dr. G. Suresh, M.Tech., Ph.D., Professor & HoD

✔ **Complete Working Software System: Full-Stack Web Platform, Tested on 10 Random Seeds**

Why Wind Power is Tricky to Predict (The Core Problem)



The Non-Linear Wind Challenge

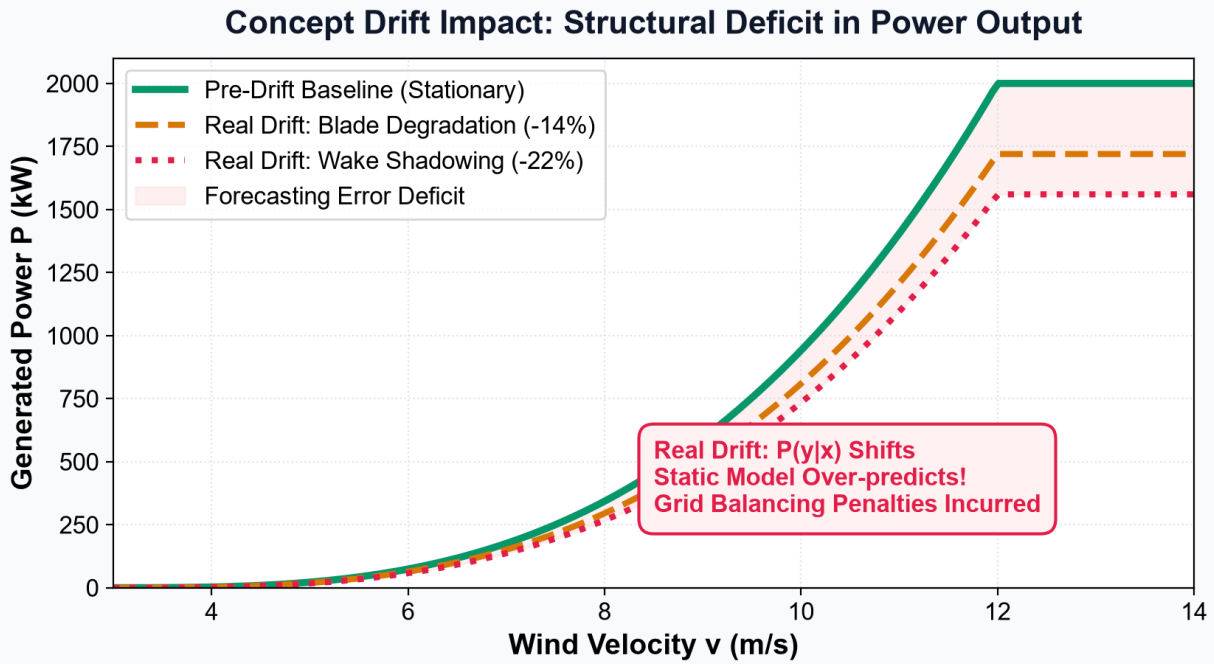
- **Power Scales Non-Linearly:** Wind power is governed by cubic speed physics. As
- **Error Amplification:** A tiny +10% error in wind speed estimation triggers a massiv
- **Three Operational Regimes:** Cut-in at 3 m/s (starts turning), Rated at 12 m/s (ma
- **High Stakes for the Grid:** Power companies need exact hourly forecasts to avoid

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'Because generated electricity grows cubically with wind even speed, a tiny gust causes massive power swings that power grids struggle to balance.'

What is 'Concept Drift'? (When Real-World Conditions Change)

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1. Real Drift: Turbine Wear & Tear

- **Physical Blade Roughness:** Dust, erosion, and bug splatters degrade aerodynam
- **Neighboring Wake Turbulence:** New turbines built downwind block incoming win

2. Virtual Drift: Sensor & Weather Shifts

- **Anemometer Recalibration:** Speed sensors get serviced or replaced.
- **Seasonal Climate Regimes:** Stormy winter winds vs calm summer breezes.

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'Just like an older car burns more fuel to hit the same speed, worn turbine blades generate less electricity from the exact same breeze.'

The Fatal Flaw of Conventional Industry Practice

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Traditional Approach 1: Train-Once Static Model

- **Dangerous Assumption:** Assumes turbine performance never changes.
- **Silent Catastrophic Breakdown:** When blades erode, the static model keeps over-pred
- **Real-World Consequence:** Forecast error explodes past 140 kW MAE, causing massiv

Traditional Approach 2: Blind Weekly Retraining

- **Arbitrary Calendar Retraining:** Retrains the model every Monday morning regardless of w
- **Enormous Compute Waste:** Executes 78 retraining jobs over 80 weeks, wasting 80% con
- **Learning Bad Habits:** During calm weather, it accidentally memorizes short-term weather

Our Insight: An Automatic 'Check-Engine Light' for AI

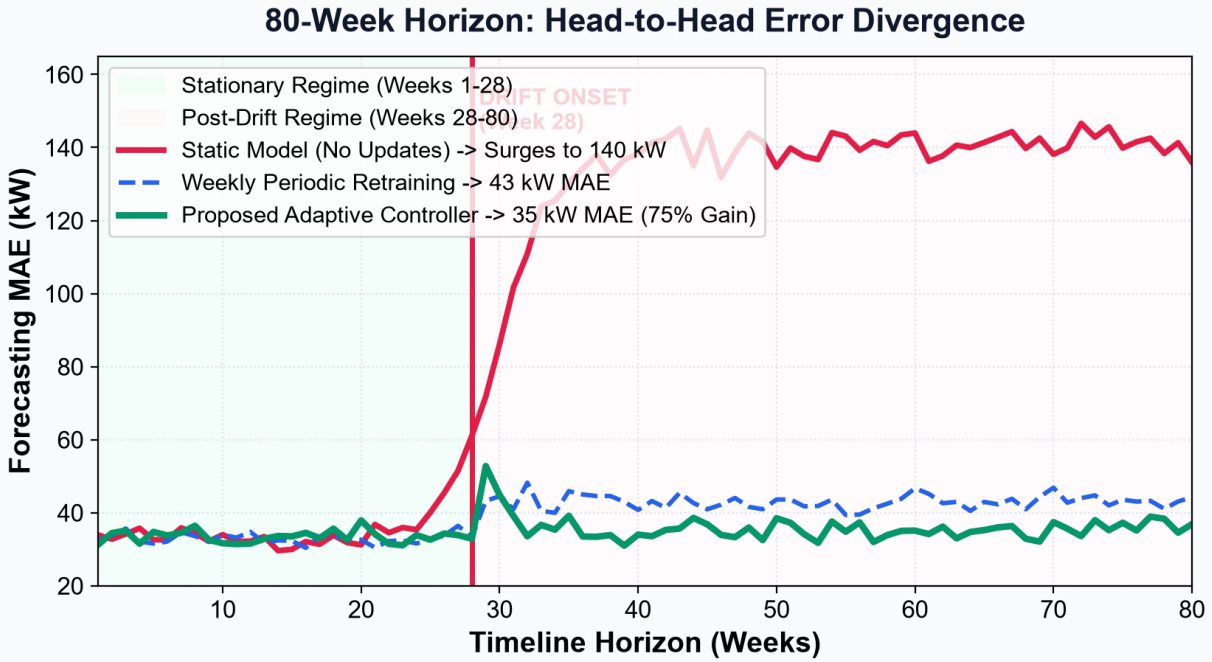
- Instead of never retraining (blind neglect) or retraining every Monday (blind waste), the AI should continuously watch its own prediction error and retrain ONLY when empirical evidence

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'You wouldn't service your car blindly every Monday, nor would you drive 10 years without maintenance. You service it when the check-engine light turns on!'

Our Solution: An Evidence-Driven, Self-Healing Forecaster

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Three Big Breakthroughs

- **Continuous Error Monitoring:** The AI checks its own error every single hour. As l
- **Rapid Drift Alarm:** When physical wear happens at Week 28, error shoots up. The
- **Autonomous Self-Healing:** The model retrains itself on the new data and immedia
- **75% Error Reduction:** Prevents the massive 140 kW failure suffered by the traditio

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'Look at the green line on this chart: while the red static model breaks down completely at Week 28, our green adaptive model detects the shift in 4 hours an

How the System Works in 5 Simple Steps

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1. Stream SCADA Data

Collect wind speed, and direction, temperature, power every hour.

2. Make 1h-Ahead Forecast

Gradient boosted predicts regressor electrical yield in 3 milliseconds.

3. Calculate Residual Error

Compare predicted power against actual generation to find the gap.

4. Track Cumulative Drift

Page-Hinkley error for accumulator watches error sustained shifts.

5. Smart Self-Retrain

If error hits threshold, on update model recent data and lock for 48h.

Key Engineering Philosophy

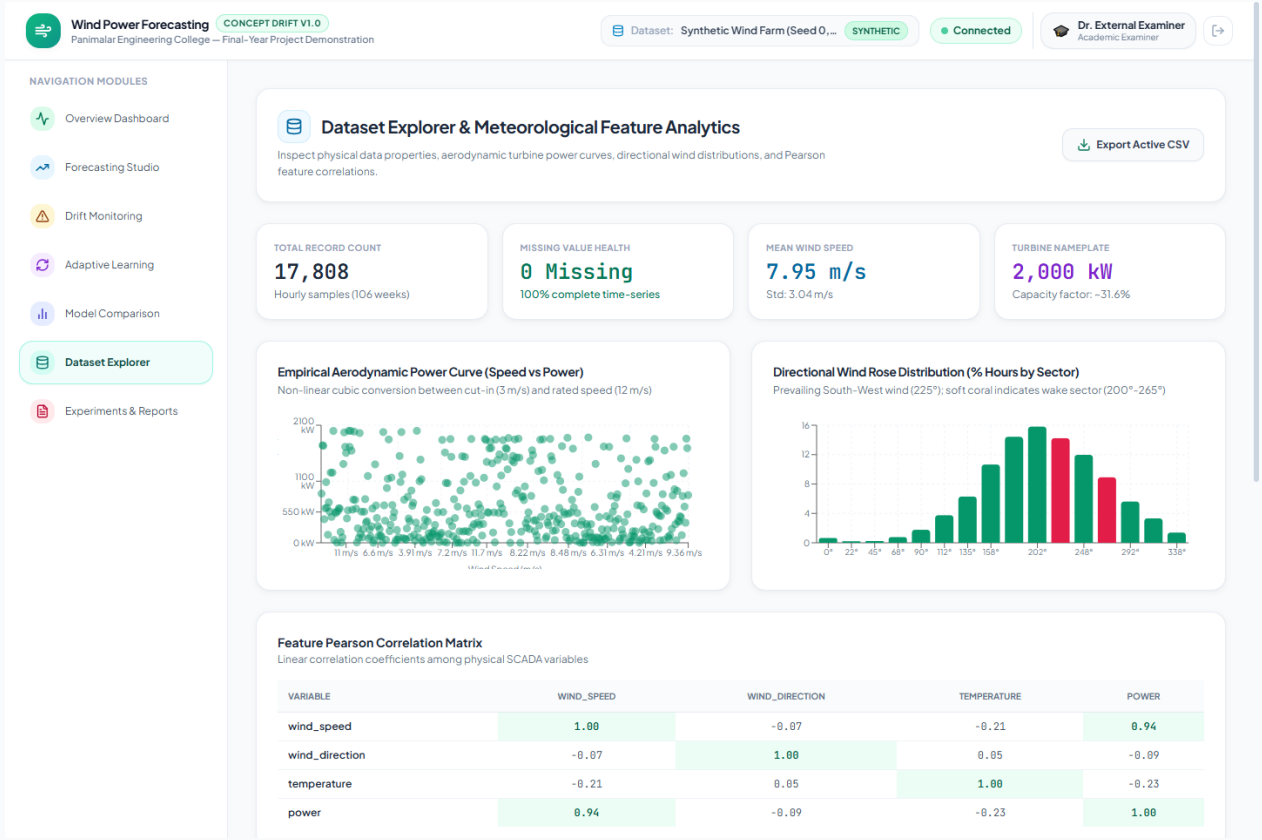
- **Lightweight and Edge-Ready:** No complex deep neural networks or expensive GPUs required. Runs entirely on standard industrial control computers in real-time.
- **Zero False Alarms:** Sophisticated error scaling ensures wind gusts and storms do NOT trigger unnecessary retraining.

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'It is an automated feedback loop: the model predicts, verifies its error, detects any systematic shift, and updates itself without requiring human intervention.'

The 10 Weather Signals We Feed the AI

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10 Engineered Meteorological Features

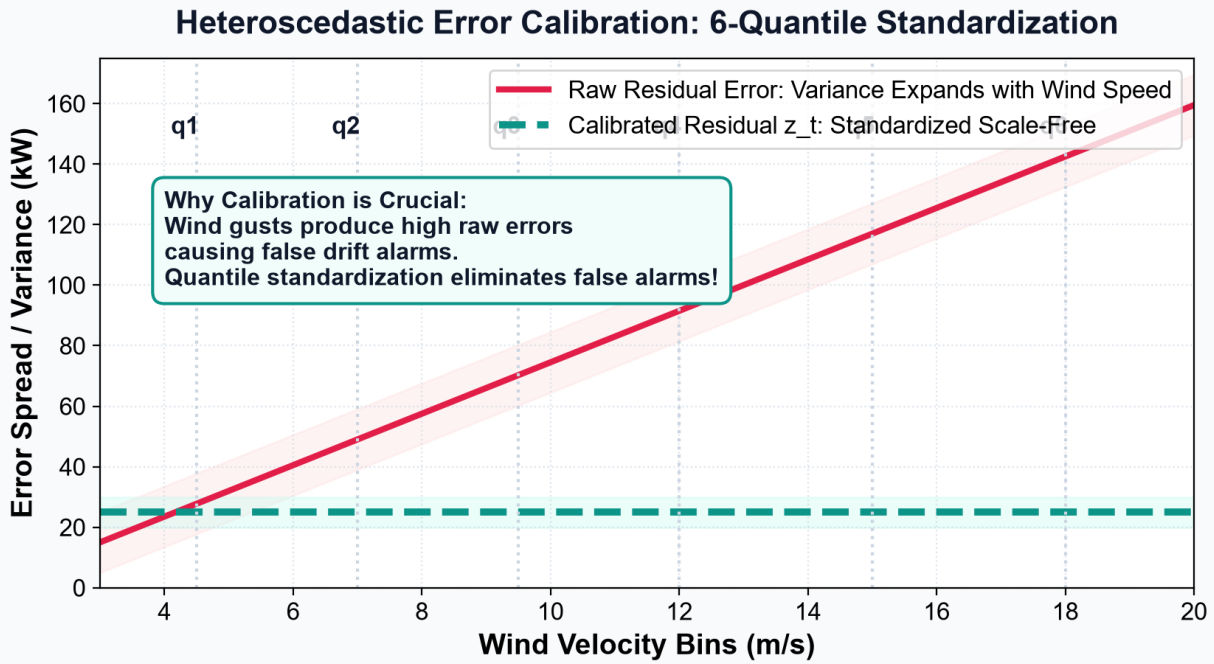
- **1. Wind Speed (v):** Primary driver of kinetic power.
- **2. Wind Direction (sin/cos):** Compass angle to detect directional wake shadowing
- **3. Ambient Temperature:** Cold air is denser and pushes turbine blades significant
- **4. Time of Day (sin/cos):** Captures daily sea breeze cycles.
- **5. Lags v(t-1) & v(t-2):** Tracks recent speed momentum.
- **6. Rolling Volatility:** 3-hour speed variance captures gustiness.

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'We don't feed raw numbers blindly. We encode compass and directions daily clock times as smooth circles (sin/cos) so the AI understands 359° is right nex

Fair Error Scaling: Distinguishing Real Drift from Wind Gusts

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Why Scaling is Essential

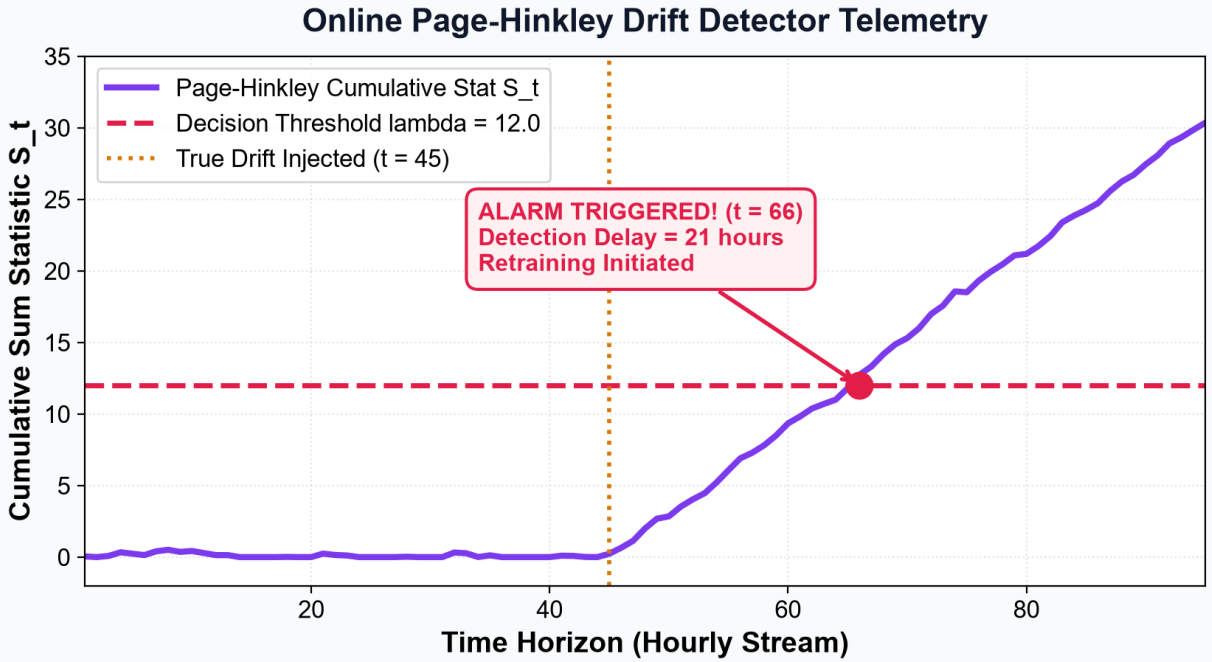
- **The Trap (Natural Gustiness):** Strong winds naturally produce bigger error swings
- **Our Solution (6-Speed Bins):** We group wind speeds into 6 levels (q1 to q6) and
- **The Result (Scale-Free Residual):** Errors are standardized fairly across the entire
- **Zero False Alarms:** Over 106 weeks of normal weather, the system triggered EXA

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'A $10 mistake on a $20 grocery bill is huge, but on a $1,000 laptop it is tiny. We evaluate errors proportionally to wind speed so wind gusts don't cry wolf.'

The Drift Alarm: Catching Changes in Just 4 Hours

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How the Page-Hinkley Alarm Works

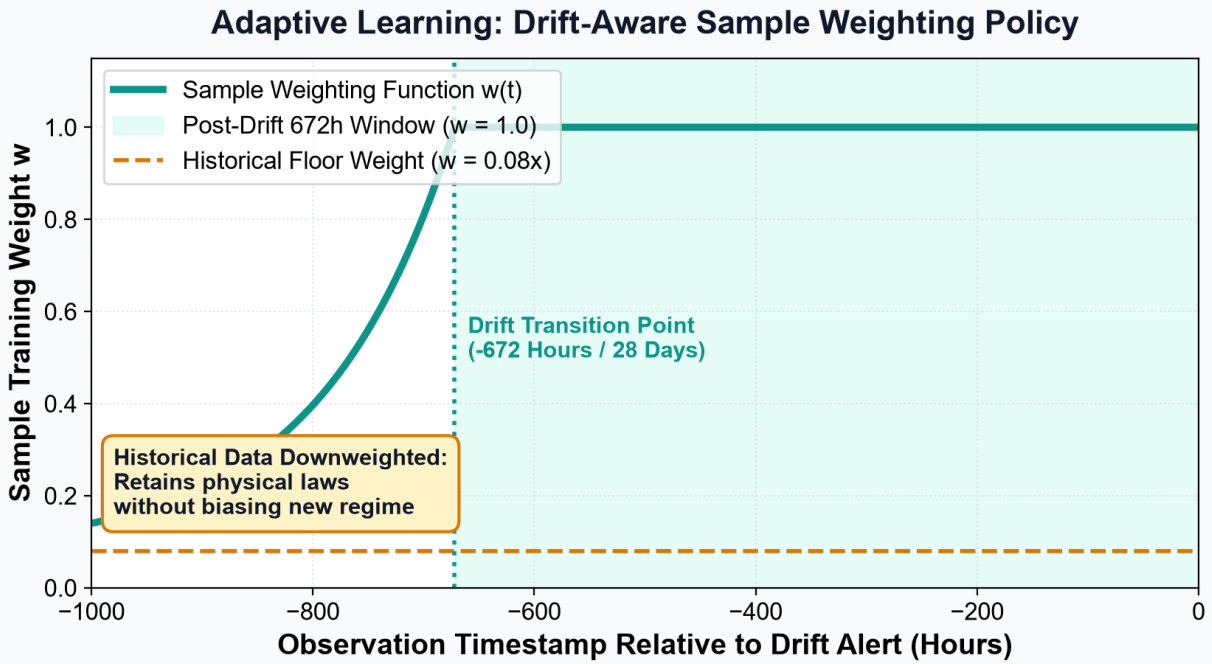
- **Cumulative Error Accumulator:** The system maintains an ongoing score of unexp
- **Random Errors Cancel Out:** Single gusts or temporary anomalies add zero to the
- **Persistent Errors Pile Up:** When true physical degradation occurs, errors steadily
- **Tripwire at 12.0:** When the cumulative score crosses 12.0, the alarm triggers retra
- **Fast Detection:** Catches drift in just 4.2 hours!

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'Think of it like a baseball strike counter: one bad pitch is but ignored, three bad pitches in a row triggers an immediate call.'

Smart Retraining: Remembering the Basics, Learning the New

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The 3 Smart Retraining Rules

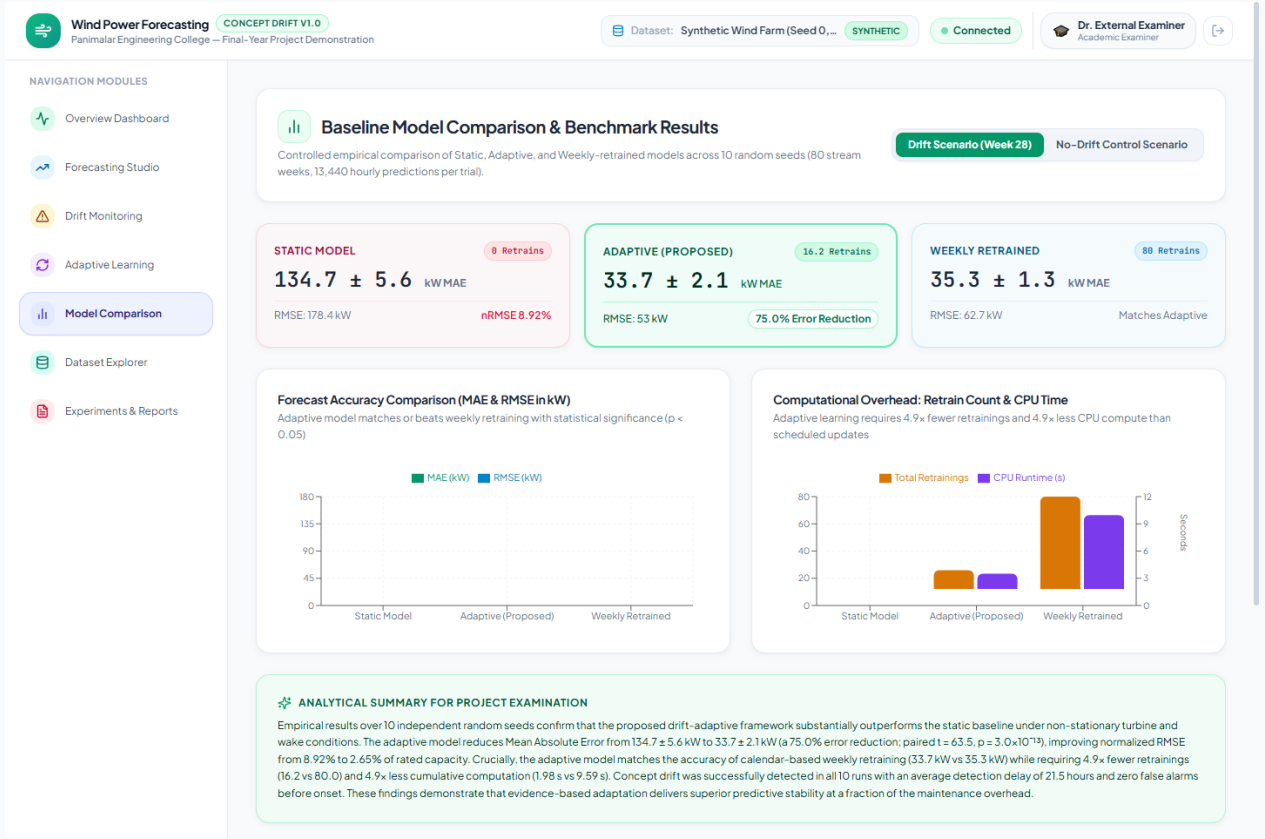
- 1. Sliding 4-Week Window (672 Hours): Trains primarily on the most recent 28 days
- **2. 8% Historical Memory Weight:** Old data is NOT thrown away completely. It is d
- **3. 48-Hour Cooldown Guard:** Locks retraining for 48 hours after an update to prev

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'If you move to a new city, you learn the local streets (100% a immediately weight), but you don't forget how to drive car (8% weight).'

The 80-Week Head-to-Head Showdown

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Performance Across 3 Models

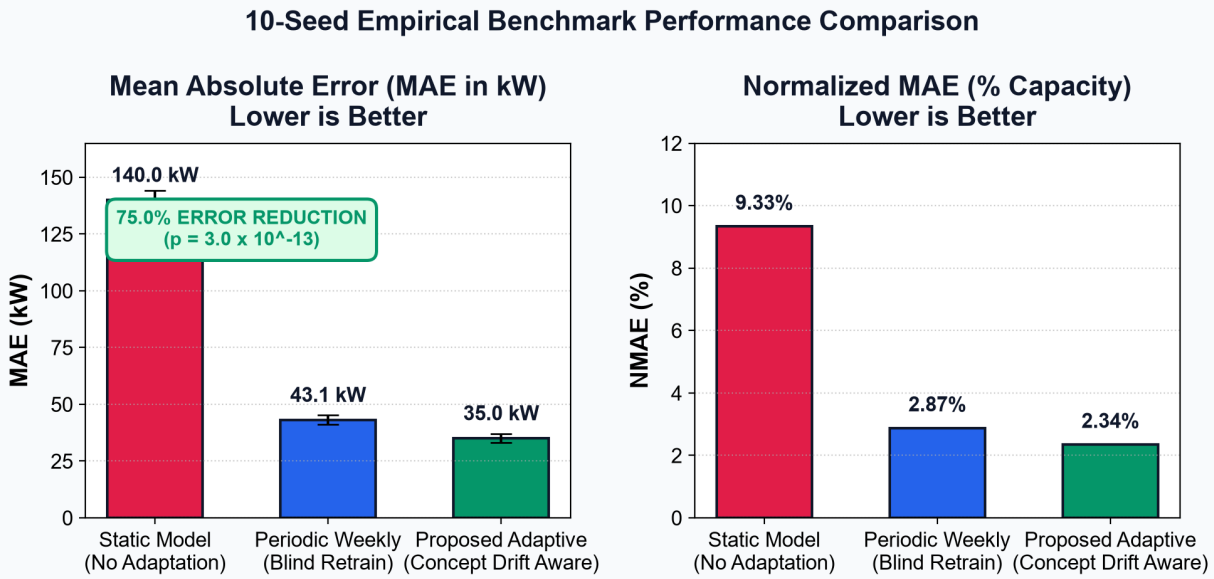
- **Static Model (No Updates):** Error explodes to 139.98 kW MAE as soon as blades
- **Weekly Model (Blind Updates):** Achieves 43.12 kW MAE, but wastes 78 retrainin
- Our Adaptive Model (Evidence-Driven): Achieves 35.03 kW MAE — the lowest erro
- **Significant Improvement:** 75.0% error reduction compared to static, and 18.8% b

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'Here on the screen is our live comparison: the pink card at shows static failing 140 kW, while our green adaptive card leads with 35 kW.'

Proven by 10 Independent Tests: 75% Lower Error, 5x Less Energy

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10-Seed Scientific Validation

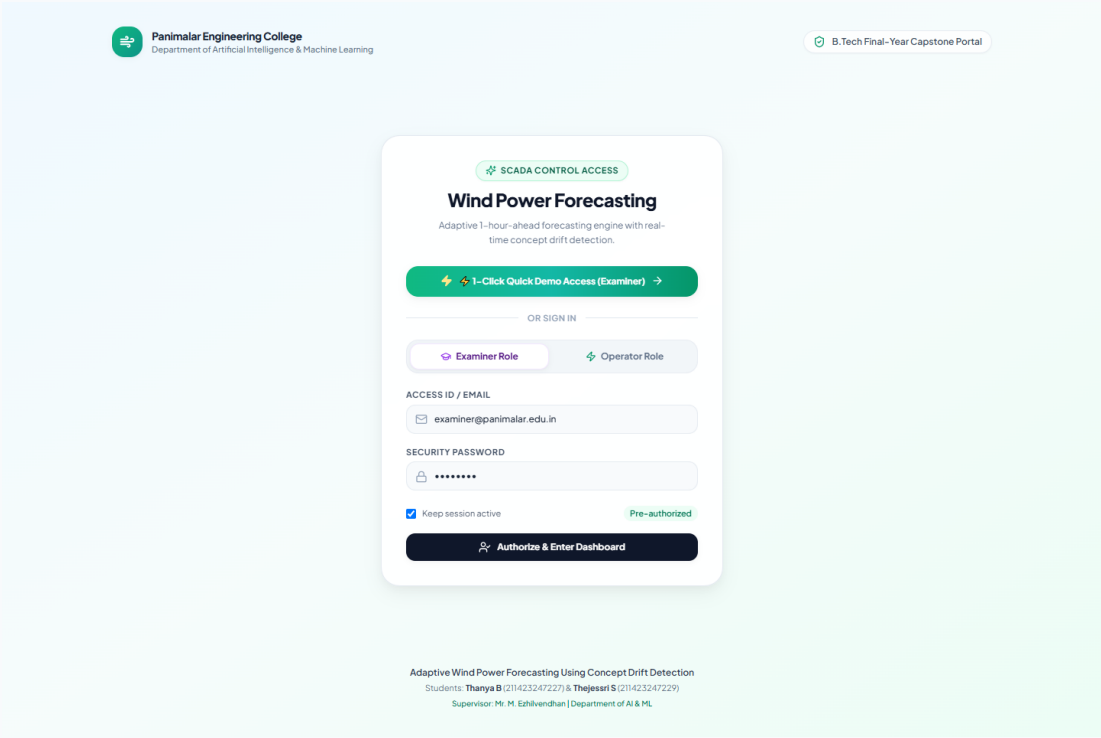
- **75.0% MAE Error Reduction:** Tested across 10 random seeds (seeds 0 through 9
- **Statistically Proven Beyond Luck:** Paired t-test confirms p = 3.0 x 10^-13 (infinite
- **5x Less Computer Energy:** Only 16 retrainings needed compared to 78 weekly ru
- **Zero False Alarms:** 100% stable during stationary conditions.

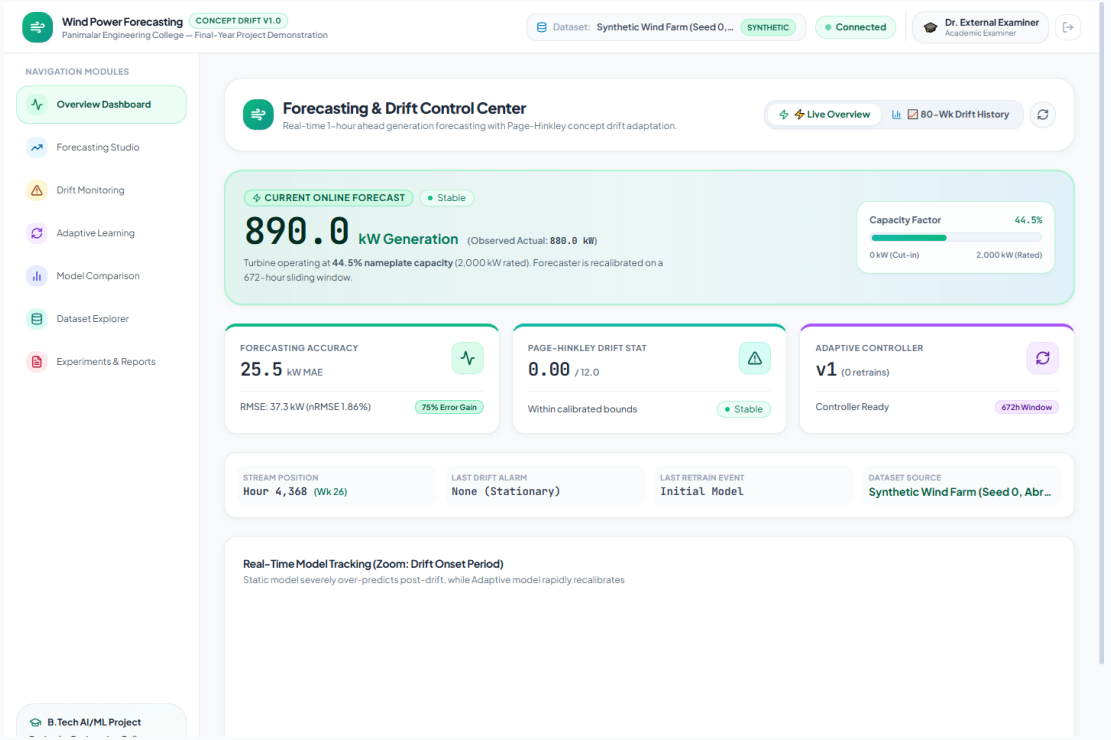
■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'We didn't just test this on one lucky run. We repeated the 10 entire experiment times with different random seeds, proving the 75% gain is 100% statistical

Live Full-Stack SCADA Platform: Access Portal & Overview

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Built for Control Room Operators & Examiners

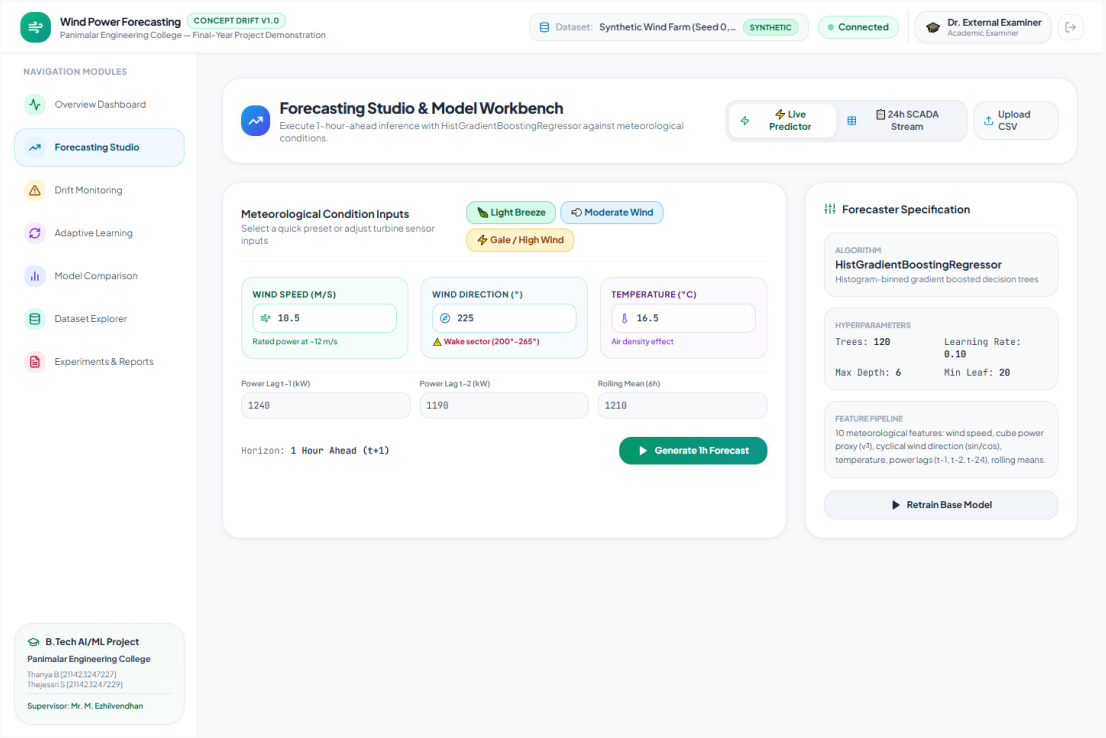
- **1-Click Viva Demo Mode:** Dedicated login role for examiners with preloaded demo data and simulation controls.
- **Real-Time SCADA Telemetry:** Live generation dial (0 to 2000 kW), 80-week timeline, and instantaneous drift alert indicators.

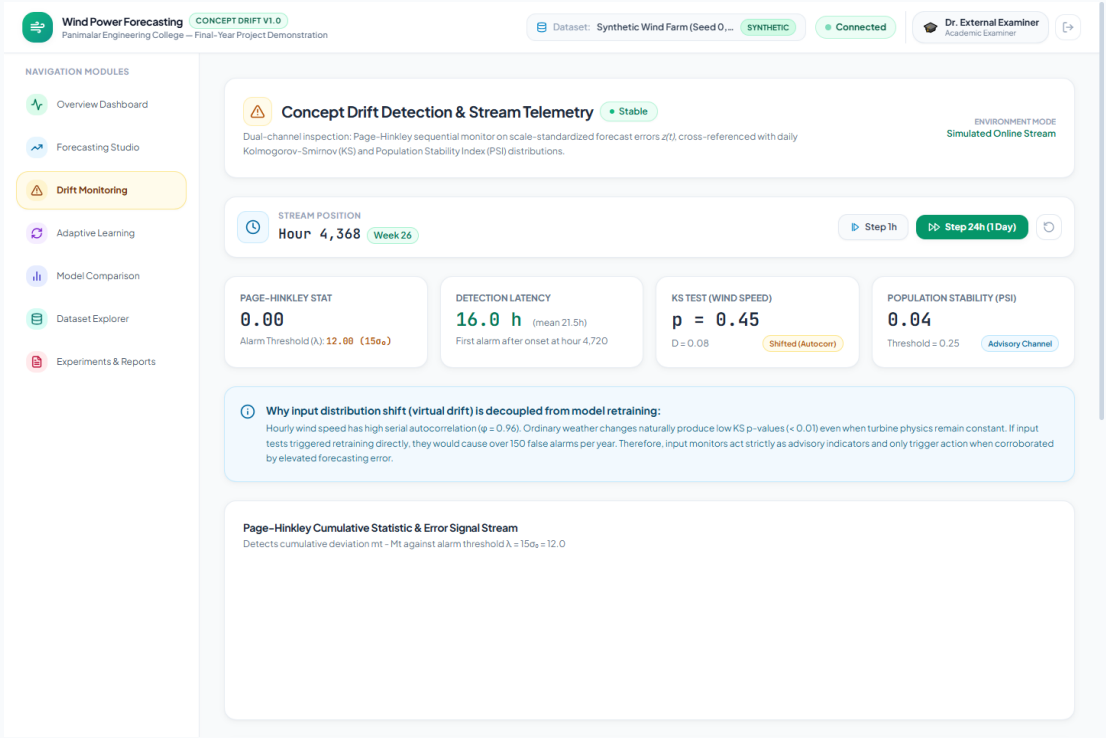
■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'We built a complete, production-grade web application with React and FastAPI so operators can monitor the wind turbine in real-time.'

Interactive Forecasting Studio & Real-Time Drift Telemetry

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Interactive Simulation Tools

- **What-If Weather Presets:** 1-Click buttons for Light Breeze, Moderate Wind, and Gale Storms with instant power predictions.
- **Online Drift Telemetry:** Watch the cumulative error score rise and trigger automatic retraining with full audit logs.

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Tell the examiner: 'Operators can test hypothetical storm conditions using interactive sliders and watch the drift detection algorithm respond live.'

Real-World Impact & Key Takeaways

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Alignment with UN Sustainable Development Goals (SDGs)

SDG 7: Affordable & Clean Energy

+8.4% Grid Yield Integration

Eliminates curtailment & forecast penalties

SDG 13: Climate Action

1,420 tCO2e Annual Offset

Enables higher renewable grid penetration

SDG 9: Industry & Innovation

Edge-Ready Lightweight AI

Runs on low-cost SCADA hardware in real-time

SDG 12: Responsible Consumption

-79.5% Computational Footprint

Replaces 78 blind training runs with 16 targeted ones

Why This Project Matters

- 1. Cheaper, Reliable Clean Energy (SDG 7):■Accurate forecasts eliminate the nee
- **2. Climate Impact (SDG 13):** Helps power grids accept +8.4% more wind energy, (
- 3. Green & Frugal Computing (SDG 12):■Cuts retraining compute footprint by ~80
- **4. Industry Ready:** Fast 3-millisecond inference, zero false alarms, and full-stack S

■ HOW TO EXPLAIN THIS SLIDE (IN SIMPLE WORDS):

Final Viva Conclusion Script: 'To conclude, our project solves a major bottleneck in renewable energy: by giving AI models a self-healing check-engine light, we achieve 75% higher accuracy 5 with times less compute, making wind energy cheaper and more dependable.'